

NORTHOLT, United Kingdom

WMO: 036720

Lat: 51.5486N

Lon: 0.4169W

Elev: 33

StdP: 100.93

Time Zone: 0.00 (EUW)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -3.8	(c) -2.4	(d) -6.8	(e) 2.1	(f) -0.1	(g) -5.4	(h) 2.4	(i) 0.5	(j) 12.9	(k) 10.1	(l) 11.4	(m) 9.7	(n) 1.5	(o) 330	(p) 0.566

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.4	28.4	18.6	26.2	17.7	24.3	17.0	19.7	26.0	18.7	24.1	17.9	22.6	3.9	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.5	12.6	21.4	16.6	11.9	20.7	15.9	11.3	19.9	56.2	25.8	53.0	24.0	50.3	22.7	25.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 10.3	(b) 9.1	(c) 8.2	(e) -6.9	(f) 31.9	(g) 1.9	(h) 2.2	(i) -8.3	(j) 33.5	(k) -9.4	(l) 34.8	(m) -10.5	(n) 36.1	(o) -11.9	(p) 37.7		
(a) 10.3	(b) 9.1	(c) 8.2	(e) -7.3	(f) 22.0	(g) 1.9	(h) 1.3	(i) -8.7	(j) 22.9	(k) -9.8	(l) 23.7	(m) -10.8	(n) 24.4	(o) -12.2	(p) 25.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	11.1	5.0	5.4	7.3	9.8	12.9	16.0	18.2	17.9	15.1	11.8	7.7	5.5		
	DBStd	5.64	3.42	3.25	3.03	3.06	3.07	2.69	2.71	2.61	2.52	3.00	3.27	3.73		
	HDD10.0	671	157	130	92	39	8	0	0	0	17	82	146			
	HDD18.3	2719	414	361	343	256	169	79	36	39	101	204	318	398		
	CDD10.0	1069	2	2	8	33	98	181	256	245	153	72	14	5		
	CDD18.3	74	0	0	0	0	2	9	33	26	4	1	0	0		
	CDH23.3	639	0	0	0	3	24	104	268	202	34	3	0	0		
	CDH26.7	152	0	0	0	0	1	23	69	54	4	1	0	0		
(14) Wind	WSAvg	3.8	4.1	4.3	4.2	3.8	4.0	3.8	3.6	3.4	3.4	3.6	3.6	3.9		
(15) Precipitation	PrecAvg	602	54	36	43	47	49	51	44	51	54	61	58	56		
	PrecMax	820	114	92	95	102	102	128	93	150	140	175	151	119		
	PrecMin	421	10	2	6	5	4	6	7	0	3	2	8	9		
	PrecStd	100	27	21	25	24	26	31	24	31	38	39	35	28		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	13.2	14.9	18.3	23.4	25.9	29.5	31.3	31.0	26.8	21.9	16.2	14.2		
		MCWB	11.2	10.6	11.0	14.3	16.6	19.3	19.9	19.8	18.0	16.7	13.9	12.1		
	2%	DB	12.3	12.7	15.5	19.7	23.5	26.2	28.5	28.0	23.8	18.9	14.9	13.1		
		MCWB	10.5	10.0	10.2	12.7	15.9	17.7	18.4	18.7	16.7	15.0	12.7	11.4		
	5%	DB	11.2	11.5	13.8	17.3	21.2	23.8	26.4	25.4	21.5	17.5	13.9	12.2		
		MCWB	9.5	9.2	9.8	11.4	14.6	16.5	17.8	17.7	15.9	14.4	12.0	10.6		
	10%	DB	10.2	10.4	12.3	15.4	18.9	21.9	24.2	23.3	20.0	16.4	12.8	11.2		
		MCWB	8.5	8.3	9.1	10.5	13.4	15.4	17.1	16.9	15.0	13.8	10.9	9.6		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	11.8	11.6	12.3	15.5	18.2	20.1	21.0	20.9	19.3	17.3	14.5	12.6		
		MCDB	12.8	13.5	15.6	21.5	23.7	27.9	28.8	29.1	23.9	20.7	15.6	13.5		
	2%	WB	10.8	10.4	11.2	13.4	16.5	18.5	19.7	19.6	18.0	16.0	13.2	11.8		
		MCDB	12.0	12.2	14.0	18.5	21.5	24.2	26.1	25.6	22.0	18.0	14.5	12.9		
	5%	WB	9.7	9.5	10.4	12.1	15.2	17.2	18.6	18.5	16.9	15.1	12.2	10.9		
		MCDB	11.0	11.2	13.0	16.1	20.0	22.3	24.3	23.5	20.0	16.8	13.6	12.1		
	10%	WB	8.8	8.6	9.5	11.0	14.1	16.2	17.7	17.7	16.0	14.3	11.1	9.7		
		MCDB	10.1	10.2	12.0	14.6	18.3	20.8	22.9	21.9	18.8	16.0	12.6	11.1		
(35) Mean Daily Temperature Range	5% DB	MDBR	5.7	6.4	7.7	9.1	9.1	9.3	9.4	9.3	9.0	7.5	6.4	5.5		
		MCDBR	5.8	7.4	10.3	12.6	12.9	13.1	13.4	12.8	11.6	8.4	6.2	5.2		
		MCWBR	5.3	5.4	6.1	6.8	6.4	6.2	5.4	5.4	5.8	5.3	5.0	4.8		
	5% WB	MCDBR	5.7	6.8	8.7	11.0	11.2	11.3	11.4	10.9	9.5	7.4	6.1	5.3		
		MCWBR	5.5	5.5	5.8	6.3	6.0	6.0	5.2	5.2	5.4	5.4	5.4	5.1		
(40) Clear-Sky Solar Irradiance	taub		0.331	0.347	0.384	0.405	0.409	0.410	0.405	0.396	0.378	0.361	0.334	0.320		
	taud		2.365	2.370	2.262	2.220	2.249	2.275	2.295	2.340	2.388	2.417	2.410	2.350		
	Ebn at Noon		672	770	803	830	845	846	843	832	805	743	664	617		
	Edh at Noon		63	82	109	127	129	127	123	112	95	77	60	55		
(44) All-Sky Solar Radiation	RadAvg		0.81	1.47	2.55	3.93	4.77	5.27	4.99	4.15	3.13	1.82	0.97	0.64		
	RadStd		0.07	0.19	0.35	0.45	0.31	0.53	0.49	0.34	0.27	0.16	0.11	0.08		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days				
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3	
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(46)
		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (83 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(47)

Nomenclature: See separate page